Unimodem Diagnostics Command Reference Specification

Microsoft Corporation

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Microsoft has contributed previous versions of this specification to public standards bodies (e.g. TIA TR-30, ITU-T Study Group 16) to support develop de jure standards for modem diagnostics. Microsoft does not know of any patents which would apply to the implementation of this specification.

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Revision History

Revision	Date	Comments
0.1	12/2/96	Initial proposal
0.2	1/22/97	Revised in response to initial industry feedback
0.3	1/27/97	Revised in response to initial industry feedback
0.4	3/3/97	Critique from MSN, Unimodem dev team
0.5	3/4/97	Critique from TIA TR-30.1 members
0.6	3/21/97	Critique from lead modem IHVs, release to WinHEC
0.7	5/1/97	Review by Unimodem dev team, ISPs, more modem IHVs
0.8	6/13/97	Second presentation to TR-30
0.85	10/15/97	Third presentation to TR-30
0.88	5/13/98	Incorporation of feedback, proposed implementation rules
0.9	6/11/98	Details on implementation requirements, testing. Open for final industry review.
1.0	7/02/98	Final release

Introduction

Unimodem™ the Universal Modem Driver, is included in Windows 95™, Windows 98™, Windows NT™ 4.0 and 5.0. In this document, Unimodem refers to any version.

Today, the overwhelming application for modems is connection to an Internet Service Provider (ISP), e.g. AOL, CompuServe, MSN, AT&T, Netcom, etc. This is such a significant business that ISP connection is overtaking G3 FAX traffic as a fraction of US telephone call connect time.

Many users are having bad experiences with their ISPs. According to 1997 ISP industry surveys, over 10% of calls into ISPs fail (for reasons other than busy), and another 5-10% fail during a call. In 1998 surveys, these numbers had improved, but there is a long ways to go. With the move to PCM modems (e.g. X2™, K56Plus™, ITU V.90), these problems are likely to get worse in the short term.

The ISP industry (ISPs, telephone companies, modem companies, communications software companies) spends a lot of effort trying to diagnose these problems, and work with the local telephone companies or server-side modem vendors when they detect anything they can actually do something about.

All of the members of this industry need to collaborate on means to identify and solve these problems. Therefore, a critical step is to define and use means for the modems to collect and report call diagnostics. These results can be used in conjunction with other standard commands and responses (e.g. connect result codes, Caller ID reports, etc). This specification leverages off of industry experience, such as the ATI6 and ATI11 commands (3Com).

Note that TIA committee TR-30 and ITU-T Study Group 16 have expressed interest to develop a modem diagnostics specification, possibly descended from this one.

Scope

This specification defines a command for use in collecting modem call diagnostics information at the end of a PSTN data call. Facsimile, voice and SVD calls may be the added in subsequent drafts. Higher layer negotiations (e.g. PPP) which are implemented above the modem are excluded.

This specification is oriented towards data calls, with a little that is applicable to fax and multimedia calls. Applicability to voice (e.g. V.253), mobile (e.g. GSM digital cellular) ISDN, ADSL (e.g. T1.413, G.lite) or Cable modems is beyond the scope of this version of this specification.

This specification does not address the syntax of results to a DTE test command, e.g. AT#UD=?.
It is recommended that modems report ERROR to that command until a specification for that syntax exists.

Normative References

ITU-T V.250 –1998, Serial Asynchronous Automatic Dialing and Control (formerly V.25 *ter*)
ITU-T V.34-1996, A modem operating at data signaling rates of up to 33600 bit/s for use on the general switched telephone network and on leased point-to-point 2-wire telephone-type circuits.
ITU-T V.58-1992, Management information model for V Series DCE's
ITU-T V.80-1996, In band DCE control and synchronous data modes for asynchronous DTE.
ITU-T (draft) V.90, A digital modem and analogue modem pair for use between a public switched telephone network (PSTN) and a digital switched network at data signalling rates of up to 56000 bit/s downstream and up to 33600 bit/s upstream.
ETS 300 001, Attachments to Public Switched Telephone Network (PSTN); General technical requirements for equipment connected to an analogue subscriber interface in the PSTN. Chapter 5: Calling Function

Data Call State Model

For purposes of this specification, there are four data call states, and associated status issues:

Call Setup:
- Calling DCE: get dial tone, generate dial digits, detect call progress signals.
- Answering DCE: detect ringing, detect CallerID, etc.

Negotiation:
- V.25 calling tone/answer tone exchanges
- V.8 or V.8bis call function negotiations
- V-series modem carrier detection and training
- modem-to-modem protocols (e.g. V.42, V.42bis).

Data Transfer:
- Bit-error rates, for each direction
- Rate renegotiation
- Retraining

Call Termination:
- protocol disconnect signals
- carrier disconnect signals
- loss of carrier
- excessive error rates

Command Syntax: AT#UD

Note: an ITU recommendation would be of the form AT+<name>, as per V.250.

Description

#UD is an action command. It does not take parameters.
It shall be the last command in the command line.

Modems conforming to this specification shall log aspects of their operation for each call, and save these results until cleared by one of the following events:

1. Power off (e.g. states D1-D3 entered)
2. Hard reset (e.g. negate DTR/ckt108.2 with &D3 set; reset button)
3. Soft reset = ATZ (Note) or AT&F
4. ATD command issued
5. ATA command issued
6. Automatic answer (e.g. set register S0>0 and ring detected)

Note: in the case that the DTE issues a soft reset *during an active call* (e.g. ATZ), it is recommended that the DTE record value 51 for key 60, to indicate that this is the reason for call termination.

These results are NOT cleared by changing DTR, V.24 circuit 108.2, if &D0, &D1 or &D2.

In response to this command, the modem shall report one or more lines of information text as defined below. Information text format is defined in ITU V.250; each line is preceded by a <CR><LF> pair, and terminated by <CR><LF>. (Note: as per V.250, CR and LF characters may be changed by writing new values to the contents of registers S2 and S3 respectively.)

The modem may generate a single line or multiple lines, based on modem capabilities and call results, followed by a standard OK final result code. For example, if call setup failed, only that result is useful. Each information text line shall be formatted as follows, including one or more key=value pairs:

DIAG <token key=value [[key=value] [key=value]] …>

Character	description
DIAG	5 characters, hex: 44, 49, 41, 47, 20
<	left angle bracket (less than sign), hex: 3C
token	unique 32-bit hexadecimal string 2A4D3263, hex: 32, 41, 34, 44, 33, 32, 36, 33
space	space character, hex 20
Key	One or two digit hexadecimal number, see Table 1
=	equal sign, hex: 3D
Value	any string as defined below (Tables 1-3, etc.)
>	right angle bracket (greater than sign), hex: 3E

Unless otherwise noted, all values are hexadecimal numbers. Any numeric values from tables in ITU V.58 are converted to hexadecimal. Multi-digit values are reported MSD first. Leading 0's may be deleted. See examples in Appendix I.

Monitoring an active connection

This command is intended for use after call termination. However, codes are defined so that a modem can respond before the first call is placed, and during a call for live monitoring purposes. For example, key 60, call termination, has value 1 defined, indicating that the call is still in progress.

There are at least two ways to do this. First, the DTE could switch the modem to On-Line command state, issue the command, capture the responses and then issue an ATO command. For smoother on-line monitoring, in-band means defined in ITU V.80 are recommended if available in the modem. If V.80 methods are used, each response line shall be a separate extended in-band message.

Implementation

This specification is based on the requirements and recommendation in the PC 98 System Design Guide, and the draft PC 99 System Design Guide. Reporting features are required based on modem features called out in the modem chapters.

Manufacturers have been prototyping implementations based on public draft versions of this specification. Therefore, implementation status is referenced to versions 0.8, 0.9 and 1.0.

Tables 1-9 each have a "status" column. The following entries are defined:
Req10 = Required for implementations of version 1.0 or later
Req09 = Required for implementations of version 0.9 or later
Req08 = Required for implementations of version 0.8 or later
ReqIF = Required only if the associated capability is supported by the DCE
Rec10 = Recommended for all implementations

Note that versions up to and including 0.8 did not include implementation requirements.

Table 1 - AT#UD Last Call Status Report Format

Key	value(s)	status	Definition
0	2 digits	Req08	Diagnostic Command Specification revision number, digit.digit
1	Table 2	Req08	Call Setup Result code
2	Table 3	Req09	Multi-media mode
3	Table 4	Req09	DTE-DCE interface mode
4	String	Rec10	V.8 CM octet string, same format as V.25ter Annex A, in quotes
5	String	Rec10	V.8 JM octet string, same format as V.25ter Annex A, in quotes
6-F			Reserved for call negotiation reports
10	0-2F	Req08	Received signal power level, in –dBm (0-43)
11	0-1F	Req08	Transmit signal power level, in –dBm (e.g. 0-17)
12	0-64	Req08	Estimated noise level, in –dBm (e.g. 10-90)
13	0-FF	Rec10	Normalized Mean Squared error, 100 (0x64) = minimum inter-symbol distance
14	0-3F	Rec10	Near echo loss, in units of dB (Note 4)
15	0-3F	Rec10	Far echo loss, in units of dB (Note 4)
16	0-3F	Rec10	Far echo delay, in units of ms (Note 4)
17	0-FFF	Rec10	Round Trip delay, in units of ms (Note 4)
18	Table 5	Rec10	V.34 INFO bit map (Note 4)
19-1F			Reserved for modulation setup and training reports (Note 5)
20	Table 6	Req08	Transmit Carrier Negotiation Result (Note 6)
21	Table 6	Req08	Receive Carrier Negotiation Result (Note 6)
22	0-1F40	Req08	Transmit Carrier symbol rate (0-8000) in symbol/s
23	0-1F40	Req08	Receive Carrier symbol rate (0-8000) in symbol/s
24	0-FA0	Rec10	Transmit Carrier frequency (0-4000) in Hz (Note 7)
25	0-FA0	Rec10	Receive Carrier frequency (0-4000) in Hz (Note 7)
26	0-FA00	Req08	Initial transmit carrier data rate (0-64000) in bit/s
27	0-FA00	Req08	Initial receive carrier data rate (0-64000) in bit/s
28-2F			reserved
30	0-FF	Rec10	Temporary carrier loss event count
31	0-FF	Rec10	Carrier Rate re-negotiation event count
32	0-FF	Req08	Carrier Retrains requested
33	0-FF	Req08	Carrier Retrain requests granted
34	0-FA00	Req08	Final transmit carrier data rate in bit/s
35	0-FA00	Req08	Final receive carrier data rate in bit/s
36-3F			reserved
40	Table 7	Req08	Protocol Negotiation Result (Note 8)
41	0-400	Req09	Error Control frame size in bytes
42	0-FF	Req09	Error control link timeouts in transmission
43	0-FF	Req09	Error control link NAKs received
44	Table 8	Req08	Compression Negotiation Result
45	0-800	Rec10	Compression dictionary size in bytes
46-4F			reserved
50	0-2	Req10	Transmit flow control: 0 = off; 1 = DC1/DC3; 2 = V.24 ckt 106/133
51	0-2	Req10	Receive flow control: 0 = off; 1 = DC1/DC3; 2 = V.24 ckt 106/133
52	0-FFFFFFFF	Req10	Transmit characters sent from DTE (Note 8)
53	0-FFFFFFFF	Req10	Received characters sent to DTE (Note 8)
54	0-FFFF	Req10	Transmit characters lost (data overrun errors from DTE) (Note 9)
55	0-FFFF	Req10	Received characters lost (data overrun errors to DTE) (Note 9)
56	0-FFFFFFFF	Req09	Transmit I- Frame count, if error control protocol running (Note 8)
57	0-FFFFFFFF	Req09	Received I-Frame count, if error control protocol running (Note 8)
58	0-FFFF	Req09	Transmit I-Frame error count, if error control protocol running (Note 9)
59	0-FFFF	Req09	Received I- Frame error count, if error control protocol running (Note 9)
5A-5F			Reserved
60	Table 9	Req08	Termination Cause
61	0-FF	ReqIF	Call Waiting event count
62-7F			Reserved for future versions of this specification
80-FF			Reserved for manufacturer proprietary keys

Notes for Tables:

1) Some operating systems cannot deal with multiple report lines in rapid succession. Modems many insert a delay (e.g. 10ms) between information text lines.

2) The code tables include values for data and fax calls. Some of the codes are applicable only to data calls (e.g. data compression), some are applicable only to call origination (e.g. busy, answering signal detection) and some are applicable only to the answering modem (e.g. calling signal detection).

3) V.58 references
Tables 1-4 are not based on V.58.
Table 5 is based on V.34.
Tables 6-9 are based on V.58.

4) Signal power and delay reports.
This specification does not mandate which signal power values to support, or the sensitivity of noise detection, or the extent to which the DCE can measure echo delay. For all of these line parameters, the DCE shall be capable of reporting these values to the extent to which it can detect or control them.

5) PCM modem parameters
This specification is being developed while V.90 is in development in ITU-T SG16. This specification covers many parameters considered valuable for diagnosing a PCM modem connection, but it is likely that future versions of this specification will add parameters based on V.90.

6) Modulation reports, Table 6
Status report Keys 20 and 21 reference Table 6. Table 6 is derived from V.58.
This specification does not mandate which carriers must be supported. For all carriers that are supported, the DCE shall be capable of reporting that modulation using the appropriate Table 6 code point.

7) Carrier Frequency has no meaning for PCM modulation. Values reported may be ignored.

8) Data transmission reports.
A continuous connection may exceed this value for characters or frames, given enough time. If these counters hit maximum value, the DCE manufacturer may roll over to 0; monitoring software in the DTE can be held responsible to monitor this information and determine significant digits beyond 32 bits. DCE manufacturers may also maintain longer counters, and report these bits.
A 24 bit character counter will overrun on a 56,000 bps data connection in about 40 minutes.
A 32 bit character counter will overrun on a 56,000 bps data connection in a bit over a week.

9) Connection error reports.
DCE may limit these error counters to 16 bits. If limited to 16 bits, these counters should limit to the maximum value. DCE manufacturers may also maintain longer counters, and report these bits.
[If a DTE that ignores flow control to the extent of generating 65,535 data overrun or underrun errors, it certainly needs some kind of maintenance. A connection with 65,535 frame errors may deserve termination.]

Keys 54 and 55 are intended to report data loss due to character overruns in either direction. The counter for transmit overrun would be incremented each time the modem receives a character from the DTE while the modem's transmit buffer is full, presumably while the modem had indicated flow off, via XOFF or negating CTS (circuit 106). The counter for receiver overrun would be incremented each time the modem receives a character from the PSTN while the modem's receive buffer is full, presumably while the modem had received a flow off indication from the DTE, via XOFF or negated RTS/RTR (circuit 133). Receive overrun could also occur if the DCE-DTE port rate was lower than the incoming data rate.

Table 2 - Call Setup Result Codes

Code	status	Definition
0	Req08	No previous call (modem log has been cleared since any previous calls)
1	Req08	No dial tone detected
2	Rec10	Reorder signal detected, network busy
3	Req08	Busy signal detected
4	Req10	No recognized signal detected (e.g. no signal, or nothing recognizable)
5	Rec10	Voice detected * if this is a voice modem (e.g. V.253) operating in voice mode (e.g. +FCLASS=8.0)
6	ReqIF	Text telephone signal detected (see V.18)
7	Req08	Data Answering signal detected (e.g. V.25 ANS, V.8 ANSam)
8	Req10	Data Calling signal detected (e.g. V.25 CT, V.8 CI)
9	Req10	Fax Answering signal detected (e.g. T.30 CED, DIS)
A	Req10	Fax Calling signal detected (e.g. T.30 CNG)
B	ReqIF	V.8bis signal detected
C-F		Reserved

Table 3 – Multimedia modes

Code	status	Definition
0	Req09	Data Only
1	Req09	FAX Only
2	ReqIF	Voice Only * if voice mode supported (e.g. V.253, IS-101)
3		VoiceView™
4		ASVD, V.61
5		ASVD, “V.34Q”
6		DSVD, Multi-Tech
7		DSVD, 1.2
8		DSVD, V.70
9	ReqIF	Video-telephony, H.324
A	ReqIF	Other V.80 call
B-F		Reserved

Table 4 – DTE-DCE modes

Code	status	Definition
0	Req09	Async data
1	Req09	V.80 transparent synchronous mode
2	Req09	V.80 framed synchronous mode
3-F		Reserved

Table 5 – V.34 INFO bit report (applicable only to V.34 or V.90 calls)

bits	status	Source bits	Definition
31-30	Rec10	INFO0 bit 20; 0	
20-29	Rec10	INFOc bits 79-88	
16-19	Rec10	INFOc bits 26-29 or 35-38 or 44-47 or 53-56- or 62-65 or 71-74	Pre-emphasis field, selected by the symbol rate chosen
12-15	Rec10	INFOa bits 26-29	
10-11	Rec10	MP bit 50; 0	
0-9	Rec10	INFOa bits 40-49	

Table 6 - gstnModulationSchemeActive from 3.7.2/V.58

Value	status	Description
0	Req10	V.17 (G3 Fax call)
1	Req10	V.21
2	Req10	V.22
3	Req10	V.22bis
4	ReqIF	V.23 Constant Carrier (1200/75)
5	ReqIF	V.23 Switched Carrier (half duplex)
6	ReqIF	V.26bis
7	ReqIF	V.26ter
8	Req10	V.27ter (G3 Fax call)
9	Req10	V.29 HD (G3 Fax call)
A	Req08	V.32 (difficult to distinguish from V.32bis)
B	Req10	V.32bis
C	Req08	V.34
D	ReqIF	V.34 HD (G3 Fax call)
E	ReqIF	V.90 Issue 1 (asymmetric)
F	ReqIF	V.90 Issue 2 (symmetric)
E-7F		Reserved (V.58)
80	ReqIF	“X2™”
81	ReqIF	“K56FLEX™”
82	ReqIF	“V.FC”
83	ReqIF	“V.32terbo”
84	Req10	Bell 212A (if modem supports B212A)
85	Req10	Bell 103 (if modem supports B103)
80-FF		Reserved for mfgs

Table 7 - errorControl Active from 3.5.2/V.58

Value	status	Description
0	Req08	Disable/none
1	Req08	V.42 LAPM
2	Req08	V.42 Alterative protocol (MNP™)
3-7F		Reserved (V.58)
80	ReqIF	MNP10™
81	ReqIF	ECP™ Enhanced Cellular Protocol
82	ReqIF	ETC™ Enhanced Throughput Cellular
82-FF		Reserved for mfgs

Table 8 - compressionActive from 3.2.2/V.58

Value	status	Description
0	Req08	None
1	Req08	V.42bis
2-7F		Reserved (V.58)
80	ReqIF	MNP5™
81-FF		Reserved for mfgs

Table 9- callCleared codes from 3.6.4/V.58-1994

callCleared: Indicates that the DCE has gone on hook and that the previously existing network connection has been cleared. These value are hex, converted from decimal in V.58.

Value	status	Description	Notes
0	Rec10	CauseUnidentified	Call setup issues
1	Req08	No Previous call	Not in V.58
2	Req09	Call is still in progress	Not in V.58
3	ReqIF	Call Waiting signal detected	Not in V.58, only if modem can detect it
4	ReqIF	Delayed	Same as value 2A, CallAttemptsLimitExceeded
A	ReqIF	NMSinitiatedDialCall	-- Network Management System
B	ReqIF	NMSinitiatedLeasedLineRestoral	
C	ReqIF	NMSinitiatedRedial	
D	ReqIF	NMSinitiatedDialDisconnect	
14	ReqIF	PowerLoss	DCE
15	ReqIF	EquipmentFailure	
16	ReqIF	FrontPanelDisconnectRequested	If there is a front panel with this control
17	ReqIF	FrontPanelLeasedLineRestoral	
18	ReqIF	AutomaticLeasedLineRestoral	
19	ReqIF	InactivityTimerExpired	
1E		cct116RestoralRequest	DTE Interface
1F	Req10	cct108isOffInhibitsDial	
20	Req10	cct108turnedOff	This is hangup with &D2
28	ReqIF	NoNumberProvided	Prohibited by some national regulations
29	ReqIF	BlacklistedNumber	
2A	ReqIF	CallAttemptsLimitExceeded	Same as “Delayed”, see ETS 300 001
2B	ReqIF	ExtensionPhoneOffHook	If extension detection supported
2C	Req08	CallSetupFailTimerExpired	e.g. S7 timeout
2D	ReqIF	IncomingCallDetected	If incoming call while sending dial command.
2E	ReqIF	LoopCurrentInterrupted	
2F	Req08	NoDialTone	
30	ReqIF	VoiceDetected	
31	ReqIF	ReorderTone	
32	ReqIF	SitTone	
33	ReqIF	EngagedTone	
34	ReqIF	LongSpaceDisconnect	And if modem program to abort on long space
3C	Req08	CarrierLost	Signal Converter
3D	Req10	TrainingFailed	
3E	Req10	NoModulationinCommon	
3F	Req08	RetrainFailed	
40	Req10	RetrainAttemptCountExceeded	
41	Req10	GstnCleardownReceived	
42	Req10	FaxDetected	If this was not a fax call attempt (note 10)
46	ReqIF	InTestMode	Test
47	ReqIF	IntrusiveSelfTestInitiated	
50	Req08	AnyKeyAbort	Call Control
51	Req08	DteHangupCommand	If ATH was used to terminate the previous call.
52	Req10	DteResetCommand	If ATZ was used to terminate the previous call.
5A	Req10	FrameReject	Error Control
5B	Req08	NoErrorControlEstablished	Error control was required
5C	Req10	ProtocolViolation	
5D	Req08	n400exceeded	
5E	Req10	NegotiationFailed	
5F	Req10	DisconnectFrameReceived	
60	Req10	SabmeFrameReceived	
64	Req10	LossOfSynchronization	Data Compression

Note 10: Fax Detection. This is simple. If a calling data modem gets ANS and V.21 Ch2 with flags from the answering modem, the calling modem should indicate fax detected. If an answering data modem goes off hook and hears CNG, the answering data modem should indicate fax detected.

Appendix I, Example Modem Response and Usage

Table I-1 - Completed Data Call, with some errors and rate retrain during the call

Modem response line	Description
DIAG <2A4D3263 0=10>	This is version 1.0
DIAG <2A4D3263 1=06 2=0 3=0>	Data Answer signal detected; Data only; Character async
DIAG <2A4D3263 5="C14513902A" 6="A145"	V.8 Call Menu indicates: V.8 Joint Menu selects:
DIAG <2A4D3263 10=1F 11=0C 12=52>	Receive level = -31dbm; transmit level = -12dbm; noise level = -82dbm
DIAG <2A4D3263 14=03 15=05 16=10>	Far end echo delay in milliseconds; Far end echo loss in dB; Near end echo loss = 16 dB
DIAG <2A4D3263 20=C 22=780 24=0C80 26=79E0 >	Transmitter: V.34 training completed; V.34 carrier frequency = 1920; V.34 symbol rate = 3200; initial transmit rate is 31200 bit/s
DIAG <2A4D3263 21=D 25=1F40 27=DAC0 >	Receiver: V.90 training completed; V.90 symbol rate = 8000; initial receive rate is 56000 bit/s
DIAG <2A4D3263 30=00 31=03 32=01 33=01>	No carrier loss events, 3 carrier rate renegotiations attempted; 1 carrier retrain requested; 1 carrier retrain granted
DIAG <2A4D3263 34=7080 35=CB20>	Final transmit rate is 28800 bit/s; final receive rate is 52000 bit/s
DIAG <2A4D3263 40=1 41=100>	LAPM negotiation completed; frame size = 256
DIAG <2A4D3263 42=0 43=0>	No error control timeout or link NAKs
DIAG <2A4D3263 44=1 45=400>	V.42bis data compression used; dictionary size = 1024
DIAG <2A4D3263 50=2 51=2>	Hardware transmit and receive flow control
DIAG <2A4D3263 52=343CC 54=0>	213964 DTE characters transmitted, w/o underrun
DIAG <2A4D3263 53=7230E6 55=47>	7483622 DTE characters received, 71 characters lost due to receive data overrun
DIAG <2A4D3263 56=29D 58=0001>	597 (decimal) frames transmitted, with 1 frame error
DIAG <2A4D3263 58=2A4B 59=0004>	10827 (decimal) frames received, with 4 frame errors
DIAG <2A4D3263 60=51>	Local PC initiated hangup

Appendix II, Compliance and Testing issues

II.1 Introduction

This specification will be used by WHQL to test modem compliance. This section addresses how this might be done.

II.2 Minimal compliance

This document has been public in some form for over a year, but the 1.0 version of this is still weeks away. The issue is: what level of compliance is essential to meet the requirements of PC 98?
As shown in Tables 1-9, there are three levels of compliance: 0.8, 0.9 and 1.0.
Req08 is intended to respect prototype implementations generated in advance of version 0.9.
The devices from key technology providers all meet this level.
Req09 & Req10 are intended to add valuable diagnostic elements, with higher priority given to those in Req09, expected in the short term.

II.2.1 Minimum functionality, no previous calls

A DCE that has not made or answered any calls since power up or reset even need only report the following:
1. Key 00, version number, e.g. 08, 09, 10
2. Key 60, Termination Cause, set to value 1 – no previous call.

II.2.2 Minimum functionality, previous data call, version 0.8

A device shall provide the following to claim any compliance with this specification
1. Key 00, version number or 08 or higher.
2. Key 01, call setup result code (Table 2)
3. Keys 10-12, signal power
4. Keys 20-23, 26, 2, carrier negotiation
5. Keys 32-35, data transfer carrier events
6. Keys 40 & 44, protocol and compression results (Tables 7 & 8)
7. Key 60, termination cause (Table 9)

II.2.3 Proposed Minimum functionality, previous data call, version 0.9

New modems should reach the level of 0.9 within 90 days of the publication of version 0.9.
These modems should add:
1. Key 2, Multimedia mode (Table 3) particularly values 0-1.
2. Key 3, DTE-DCE interface mode
3. Keys 41-43, V.42 Error control characteristics
4. Keys 56-59, V.42 Error control operation results

II.2.4 Proposed minimum functionality, previous data call, version 1.0

New modems should reach the level of 1.0 within 6 months of the publication of 1.0.
These modems should add:
1. Keys 4-5, V.8 CM and JM bits
2. Keys 13-18, 24-25, signal converter detail
3. Keys 30-31, signal converter connection results
4. Key 45, compression dictionary size
5. Keys 50-55, port flow control diagnostics

II.3 Testing Issues

There are two elements in determining compliance to this specification:
1. Does the modem provide the required information?
2. Does the modem provide sufficiently accurate information?

Section II.2 above address the first question, defining information to provide.
The following addresses how to check the accuracy.

II.3.1 Modem choices

Key 00, the version number, is reported by the modem; there is nothing to check.

II.3.2 Local elements

Several elements are local to the DTE and DCE, not exposed to the remote system. These include:
1. Key 01, Call setup result, particularly if the remote system was not reached.
2. Key 02, Multimedia modes, since they may be set locally.
3. Key 03, DTE-DCE interface mode
4. Keys 50-51, Local port flow control (set by +IFC or equivalent commands)
5. Keys 52-53, local port character traffic
6. Keys 54-55, local port errors

Some Key 01 reports can be checked by comparison to call progress reports issued at the beginning of the call (e.g. Connect, Busy, No Dial Tone). Others depend on negotiation results.

The DTE-DCE port is a vulnerable point. The verify functionality, a test program has to deliberately induce transmit overrun errors and receive overrun errors, and then verify that they were detected by the modem.
Recommended methodology is to generate a connection at relatively low speed, transmit characters at a rate higher than the line rate while ignoring circuit 106 and to post circuit

II.3.2 Negotiation elements

A large number of elements concern negotiation results. The local and the remote DCE should agree on these elements. If they don't agree, either the diagnostic reporting is flawed, or at least one of the modems is broken! For example, they should agree on multimedia mode, modulation characteristics, error control and data compression, or data can't be transferred intact.

Other elements represent measurements. For example, different modems may come up with different measurements of estimated noise power. The modem under test will report its transmit power, but this will not match the measured receive power at the reference modem. [The difference is of diagnostic value.] Therefore, software that tests complain should not expect these values to match.

II.3.3 Data transfer elements

If the link is working correctly, the characters one modem transmits should closely match the characters the receiving modem gets, and vice versa; otherwise, there is a problem to diagnose. The first test for this is to conduct a data transfer session with minimal impairments, to verify that each modem can count. The second test is to conduct a data transfer with significant impairments, so that errors are encountered and counted, so that reports are generated. The transmitted frame count may exceed the sum of received good frames plus received bad frames, because frame errors may obscure frame boundries.

Reporting is also collaborative between the two modems. For example, transmit frame errors, key 58, is dependent on the feedback from the receiving modem. Transmit frame errors should correspond to the number of frames that have to be retransmitted due to these reports. V.250 +MS command is required

on both sides to test key 3E, no modulation in common. It may also be difficult to force retrains, in order to test Key 3F.

II.3.4 Call termination code points

This testing can be complex, due to the difficulty of generating all the termination causes.
For example, if a modem is under test, it is easy to automatically generate local call termination, and straightforward to generate normal remote termination (e.g. Key 5F, Disconnection Frame Received). However, tooling to deliberately generate some causes would require a specially built and instrumented LAPM implementation which could be programmed to generate conditions like protocol violation (key 5C) or SABME Frame Received (Key 60).

Testing must include the following, because they can be generated with common modems and commands:

1. Keys 1-2
2. Keys 1F-20
3. Keys 2C, 2F, 3C
4. Keys 50-52
5. Key 5B

Appendix III, commentary on the value of individual diagnostic elements.

All of the elements in this modem diagnostic specification are considered valuable for diagnosing connection problems. This section is an informative explanation for some of these elements.

III.1 Line conditions

The classic tech support explanation of modem problems is “line noise”. In some cases, this is still true. Keys 10-8 are directed at line conditions. Key 18 is particularly valuable for V.34 connections, because it encodes what the modems think about the line.

III.2 Carrier and protocol negotiation

The modems are most vulnerable to line conditions, and changing line conditions, in negotiation and training. Disparities between the two modems are fatal.

This should be straightforward, but there is abundant anecdotal evidence (particularly from ISPs) that there are problems and bugs here. Some of these have been fixed in the last year or so, and this may be partial cause why average connection failure rates have dropped from 16.2% to 8.9% in the last year.

III.3 Carrier and protocol operation

Retrains and rate renegotiations are common on V.34 links. So are unexplained call disconnection events. Catching these events is a big value for the diagnostics specification.

There is anecdotal evidence of bugs in the LAPM implementations of some modems that cause random errors and disconnection. Keys 5A-64 are intended to catch these problems.

III.4 Local port problems

At 115,200 bps, the electrical characteristics of serial ports are stressed. In EIA/TIA-232-F, circuits with the classic +/-12v swing are not recommended at speeds over 20,000 bps. Using the lower voltage electrical characteristics in EIA/TIA-562 (+/-5v), these are still not recommended to run over 60,000 bps. Therefore, serial port data errors are possible.

Also, in a busy computer operating system, loss of data is probable if the UART is unbuffered (16450), because the interrupt response required is under 100 microseconds per character. The buffers in the common 16550 mitigate this, but it is still a risk.

Also, data can be lost if the internal bus used to connect to the port inside the PC are tied up by other drivers. For example, PCI video cards are notorious for monopolizing the PCI bus for long periods, locking other drivers out from servicing ports (e.g. serial or USB).

Data is lost if the transmitting PC sends data too long after the modem has negated circuit 106 (Clear to Send) or sent an XOFF. Data is lost if the receiving modem fills its buffers while the PC has negated circuit 133 (Ready to Receive, same lead as Request to Send) or has sent an XOFF.

Also, if the receiving application gets behind, it may not even be directly aware of lost data. A comparison of the received data count in that application (e.g. Microsoft RAS) and the result from Key 53 (received characters sent to DTE) may illuminate problems.

III.5 Call termination events

There are a variety of events that can terminate a call. Many of them are causes most modems are not equipped to detect. There is anecdotal evidence that many users do not understand that Call Waiting signals can interrupt their modem connection. (Note: Windows TAPI has standard provisions for sending

Call Waiting disabling DTMF sequences for data calls, but they are often ignored.) In a home, somebody picking up another extension and trying to dial is another common one.